

ROHAN MATHUR

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| rm0929.github.io/developer-portfolio

EDUCATION

Arizona State University

August 2023 – May 2025

Master of Science in Software Engineering (CGPA: 3.84)

Tempe, Arizona

NMIMS University

July 2019 – May 2023

Bachelor of Technology in Computer Engineering (CGPA: 3.75)

Mumbai, India

TECHNICAL SKILLS

Programming and Backend: Java, Python, Node.js, Spring Boot, Express.js, REST APIs, Microservices, SQL, MongoDB, PostgreSQL, GraphDB

Frontend: React.js, React Native, TypeScript, JavaScript, Next.js, Flutter, HTML5, CSS3

AI/ML Integration: TensorFlow, scikit-learn, Predictive Analytics, Data Pipelines, Model Deployment

Cloud & DevOps: AWS (EC2), Docker, Kubernetes, CI/CD, Git, Jenkins, Logging & Monitoring

Practices: Agile/Scrum, OOP, Data Structures & Algorithms, Linux, Operating Systems

CERTIFICATIONS

- AWS Certified Cloud Solutions Architect - Associate
- IBM Certified Full Stack Software Developer - Professional

PROFESSIONAL EXPERIENCE

Software Engineering Intern

May 2022 – October 2022

Indian Oil Corporation Limited

Vadodara, India

- Developed **full-stack** software with **Node.js/Express** and **MongoDB** to manage and process live telemetry from refinery machines, ensuring scalability and system reliability under heavy loads.
- Integrated **AI/ML** models (**TensorFlow**, **scikit-learn**) into backend services to predict machine failures from historical sensor data, achieving **94.7% accuracy** and enabling proactive maintenance.
- Built a real-time **React** dashboard connected to the backend APIs to visualize machine health, failure risks, and performance trends for refinery operators.
- Optimized **data pipelines** with indexing, caching, and monitoring, reducing downtime by **17%** and improving query response times for live analytics.

Full Stack Developer Intern

April 2021 – September 2021

Naturesmith Foods

Delhi, India

- Architected and developed **scalable inventory management** APIs using **Node.js** and **Express** for high-traffic e-commerce platform, **improving data fetch performance by 25%**.
- Built responsive React frontend components integrated with backend APIs to manage real-time inventory tracking.
- Implemented database **query optimization** and **caching** mechanisms to reduce DB load and improve response times.
- Deployed production-ready applications to **AWS EC2** with comprehensive logging and performance tracking.

ACADEMIC PROJECTS

SENSAI - A Privacy Mobile Application | Flutter, UWB, Node.js, MongoDB, AWS

February 2025

- Developed **cross-platform mobile application** using **Flutter** to detect privacy-invading sensors, leveraging Qorvo's Ultra-wide beacon technology.
- Architected secure, **scalable backend** infrastructure using Node.js and MongoDB on AWS, handling real-time beacon data processing and push notifications.
- Implemented advanced data visualization features displaying sensor proximity, direction, and mitigation strategies.

Crime Analysis Dashboard using Web Semantics | React, Node.js, Vite, GraphDB

September 2024

- Built a **distributed** analytics platform processing 500k+ crime records, enabling scalable and efficient queries.
- Developed a **Node.js, GraphQL microservices backend**, improving accessibility and reducing query times by **60%**.
- Implemented automated ETL pipelines into **GraphDB/relational databases**, ensuring consistency and reliability.
- Leveraged **React.js** for visualization, with emphasis on backend query optimization and service orchestration.

Agile Scrum Simulator Application | Spring Boot, JavaFX, MySQL, JDBC

November 2024

- Built **Java** application using **Spring Boot** framework simulating agile development cycles with interactive GUI.
- Designed robust database architecture using **MySQL** with **JDBC** for task state tracking and user authentication.
- Integrated real-time collaboration features using **WebSockets**, **increasing user engagement by 30%**.

Glaucoma Detection using Artificial Intelligence | Python, TensorFlow, PyTorch, CNN

March 2023

- Built /ML pipeline on 20K+ images with optimized CNN architecture; deployed web app for real-time predictions.